

Notch and Catch: Dynamic Inefficiency in the Corporate Income Tax Schedule

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Abstract

Notched corporate income tax schedules are prevalent worldwide, yet their effects on dynamic allocative efficiency remain understudied. Leveraging detailed administrative employer-employee linked tax return data from Taiwan, alongside a significant reform shifting the exemption threshold, we analyze the dynamic consequences of corporate tax notches. We document substantial bunching in the corporate income distribution, as firms strategically respond by reducing real output and shifting profits toward labor salaries to position themselves just below the threshold. Moreover, we identify a “notch trap”: firms that bunch below the exemption threshold become persistently constrained, experiencing 20% lower average revenue growth and an 8.1% higher exit rate relative to firms just above the threshold.

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1 Introduction

In recent years, reducing the corporate tax burden through decreasing marginal tax rates and restricting the tax base has become a global trend as governments seek to stimulate firm production and overall economic growth. Advocates argue that a reduced corporate tax burden encourages investment and boosts production, while critics contend that these benefits have limited impact on economic expansion and come with significant fiscal revenue losses. While a growing body of research has examined firms' responses to marginal tax rate reductions, much less is known about the impacts of tax notches on firm dynamics and market efficiency.

Tax notch, where tax liability shifts abruptly at specific taxable income thresholds, are a common feature in corporate income tax systems worldwide. These thresholds often serve to foster the growth of small and medium-sized firms while ensuring larger firms contribute their fair share to tax revenues. Despite their prevalence, most research on tax notches has focused on firms' immediate, static responses near these thresholds, leaving the impacts on dynamic firm behavior largely unexplored. Understanding firms' adjustments in response to tax notches, both in terms of revenue growth (intensive margin) and enter/exit decision (extensive margin), could reveal how such policies shape broader market efficiency and composition. If firms respond to tax notches by distorting their production growth to avoid crossing thresholds, these adjustments could ultimately affect firm exit rates, market turnover, and aggregate economic performance.

This paper explores the dynamic consequences of tax notches in the corporate income tax design. We exploit a major tax reform in Taiwan in 2010, which aimed at boosting the economy and alleviating the tax burdens on small and medium-sized enterprises. It substantially raised the taxable income exemption threshold from 50 thousand NTD to 120 thousand NTD (approximately 17 thousand USD to 40 thousand USD), allowing more small firms to be exempt from corporate income tax. We utilize comprehensive administrative data sets covering corporate income tax and value-added tax filings, personal income tax records, and household registration records. Leveraging both annual corporate income tax records and third-party-reported value-added tax records, we distinguish firms' real responses from their reporting responses. Supplementary employee-employer data with kinship information further enables us to document clear misreporting channels within family networks. Moreover,

the extended panel structure allows us to examine firms' dynamic responses over time.

Our empirical analysis proceeds in two steps. In the first step, we document how firms respond to tax notches. We find that firms exhibit pronounced bunching patterns in their taxable income distribution at the exemption thresholds before the 2010 corporate income tax reform. After the reform, firms that were previously bunched below the previous exemption threshold (50 thousand NTD) inflated their taxable incomes to bunch below the new exemption threshold (120 thousand NTD). This change is immediate, and common across different industries.

How do firms achieve bunching? We demonstrate two primary channels underlying the bunching behavior: real responses by shrinking production size, and misreporting by shifting profits as labor expenditures. To separate these two channels, we analyze changes in the average shares of intermediate input costs and labor salaries in gross revenues across taxable income bins before and after the reform. We rely on both cross-sectional variations induced by the exemption thresholds and the changes in the exemption thresholds by the 2010 corporate income tax reform to examine firm responses qualitatively.

Specifically, we find that firms with taxable incomes just below the exemption threshold have discontinuously smaller intermediate input per unit of gross revenues, both before and after the reform. This pattern suggests that firms below the exemption threshold tend to scale down production by utilizing less intermediate input. On the contrary, firms with taxable incomes just below the exemption threshold have discontinuously larger labor salary per unit of gross revenues. Moreover, exploiting rich information in our employee-employer linked records with family registration, we find that firms mainly shift corporate profits as personal incomes paid to relatives of firm owners. This suggests that firms exploit relatively lower personal income tax rates to avoid paying higher corporate income taxes and reduce the overall tax burdens. Through a battery of robustness checks, we rule out alternative channels for firm bunching such as revenue under-reporting and ownership restructuring.

We next examine the dynamic consequences of firm bunching. By comparing firms with taxable income immediately below and above the exemption thresholds, we find that those bunching firms tend to have a lower revenue growth rate and a higher exit rate. Those firms which bunch below the exemption thresholds have shrunk their operation to be lower than the optimal level in the absence of tax incentives. They require extra favorable growth opportunities to justify growing beyond the exemption thresholds, but will become more

vulnerable to adverse cost shocks and are more likely to be eliminated from the market. Therefore, we uncover evidence of a “bunching trap” created by tax notches, where firms become persistently stagnant and eventually exit.

In the second step of our analysis, to further boost the confidence in empirical findings, we complement the bunching-based analyses with a regression framework, which relies on an exogenous shift in the relative distance between firms’ taxable incomes and the exemption threshold due to the 2010 corporate income tax reform. We include a rich set of fixed effects and firm size controls in the regression to account for other potential confounding factors. Our identification relies on the assumption that, conditional on firms’ taxable income, the variation in relative distance to the exemption threshold arises exogenously from policy changes and shifts firms’ incentives to bunch. We find that conditional on firm size, industry-specific macro shocks, and other firm-level control variables, firms whose taxable incomes fall below the exemption threshold due to the 2010 tax reform tend to reduce intermediate input share in gross revenue by 2% and increase the salary shares in gross revenue by 7%. Their growth rate is 2.1 percentage points lower, which is approximately 20% of the sample mean. Moreover, their exit rate is 0.7 percentage points higher and approximately 8.1% of the sample mean. These empirical findings are fully consistent with our bunching-based analyses.

To quantify the welfare consequences from the tax notches, evaluate winners and losers from the tax reform, and explore alternative designs of the tax schedule for various policy goals, we build an industry dynamics model incorporating firms’ decision on static production and tax reporting, as well as dynamic capital adjustment and exit. At the beginning of each year, firms are endowed with heterogeneous capital and productivity. They earn static profits but can potentially avoid corporate income taxes by changing firm production or misreporting labor salary. They will then decide whether to exit, or how much to invest if they decide to stay in operation. For the next steps of this project, we aim to estimate the model’s key parameters and evaluate various policy counterfactuals.

Our paper contributes to the following two strands of literature. First, this paper contributes to a substantial body of work analyzing firms’ behavioral responses to the notched corporate income tax schedule. The literature generally focuses on estimating the implied elasticity of corporate taxable income from static bunching patterns ([Auerbach, Hines Jr, and Slemrod, 2007](#); [Dwenger and Steiner, 2012](#); [Bachas and Soto, 2021](#)) or differentiating

specific real and reporting responses in a static setting (Devereux, Liu, and Loretz, 2014; Coles, Patel, Seegert, and Smith, 2022; Lobel, Scot, and Zúniga, 2024).¹ Our paper provides firm-level evidence on firms’ dynamic growth, entry, exit, and productivity, in addition to their static responses, and distinguishes between real and reporting strategies. Moreover, there is a growing literature on firms’ dynamic responses to size-based thresholds (Tam, 2023; Choudhary and Gupta, 2024; Garbinti, Goupille-Lebret, Muñoz, Stantcheva, and Zucman, 2023; Muthitacharoen, Wanichthaworn, and Burong, 2021), which mainly focuses on value-added tax thresholds defined in terms of gross revenue. We extend this by documenting firms’ dynamic responses to the notched corporate income tax schedule.

Second, our paper relates to the broader literature examining the effects of regulations and reforms on market efficiency and dynamics, including R&D incentives (Chen, Liu, Suárez Serrato, and Xu, 2021) and labor regulations (Braguinsky, Branstetter, and Regateiro, 2011; Gourio and Roys, 2014; Garicano, Lelarge, and Van Reenen, 2016; Amirapu and Gechter, 2020). Our policy simulations aim to contribute to this understanding by analyzing the aggregate equity and efficiency implications of corporate income tax notches (Swonder and Vergara, 2025). Additionally, our study offers new perspective on “missing middle” size distribution patterns (Dharmapala, Slemrod, and Wilson, 2011) and the origins of small and medium-sized enterprises (SMEs) through the lens of tax policy. For example, Aw, Chung, and Roberts (2003) empirically highlights that Taiwanese manufacturers experience a higher turnover rate compared to those in South Korea, underscoring unique dynamics in SME formation.

The remainder of this paper is structured as follows. Section 2 describes the overall corporate taxation in Taiwan and the corporate income tax reform in 2010. Section 3 introduces the data used for our empirical analysis. Section 4 presents bunching-based analyses on how firms respond to tax notches, followed up by the regression analysis exploiting the 2010 tax reform. Section 5 develop the model of firms with notched corporate income tax schedule. Finally, Section 6 concludes the paper for now.

¹Devereux, Liu, and Loretz (2014) explore the option for owner-managers to over-report deductible expenses. Bachas and Soto (2021) decompose the elasticity of corporate taxable income into elasticity of revenue and deduction costs, using a unique corporate tax schedule based on gross revenue in Costa Rica.

2 Policy Background

2.1 Corporate Taxation in Taiwan

Corporate income tax has been the largest source of gross tax revenues in Taiwan. Over the period 2000–2024, Taiwan’s annual gross tax revenue averaged 2.1 trillion NTD, of which corporate income tax contributed 23%, personal income tax contributed 20%, and value-added tax contributed 16% (Appendix Figure A.1). All profit-seeking enterprises operating within Taiwan are potentially subject to the corporate income tax.² Corporate income tax is assessed annually on firms’ gross profits, which are calculated as total revenues net of deductible costs, including expenditures on labor, materials, capital, and other qualifying business expenses. The standard corporate income tax schedule is progressive with notches: firms with annual taxable income below the threshold face a zero statutory tax rate; Once the taxable income exceeds the threshold, the statutory tax rate is applied to the entire taxable income, resulting in a discrete jump in total tax liability.

All corporations are required to file corporate income tax returns annually before the end of May for the preceding fiscal year, even if they are temporarily closed or undergo negative taxable income.³ To ensure proper filing and compliance with corporate income tax regulations, Taiwan’s tax authority conducts random audits and onsite inspections, particularly to detect unregistered firms, late filings, or delayed payments. The consequences of such violations can be severe. Businesses operating failing to submit timely annual tax reports may face fines, criminal charges, and civil liabilities. Non-compliance is monitored by the Ministry of Finance and may result in further financial penalties or criminal prosecution.⁴

Another important tax faced by profit-seeking enterprises is the value-added tax (VAT). Under the VAT system, a standard rate of 5% is levied on the value added at each stage

²Sole proprietorship and partnership only file the corporate income tax but are not requirement to any payment, as their income are subject to the personal income tax at the individual shareholder level. Our paper exclude them from all analyses.

³Firms which meet eligibility criteria based on their gross revenues and profits can opt for simplified filing methods called the Expanded Paper Review. Due to a different tax schedule as well as low adoption rate, we exclude firms under the Expanded Paper Review from the analyses.

⁴Late payments are subject to a 1% penalty for every three days of delay, up to a maximum of 30 days. Beyond that period, daily interest is charged according to the postal savings one-year fixed deposit rate. Please refer to the official documents from Taiwan Ministry of Finance accessed at <https://www.etax.nat.gov.tw/etwmain/tax-info/understanding/tax-q-and-a/national/profit-seeking-enterprise-income-tax/penalty/2J251pa>. Under the Article 110 of the Income Tax Act accessed at <https://law.moj.gov.tw/ENG/LawClass/LawAll.aspx?pcode=G0340003>, the penalty for corporate income tax evasion can be up to three times the amount of tax evaded.

of production and distribution, calculated as the difference between a firm’s revenue and deductible input purchases. Firms are required to record and file transaction-level VAT returns every two months, which enables tax authorities to trace detailed business-to-business transactions including materials and capital input purchases. It ensures firms to accurately report and validate their sales and purchases to receive VAT credits.⁵

2.2 The 2010 Corporate Income Tax Reform

Taiwan tax authority implemented one of the largest income tax reform in 2010, aimed at boosting the economy and alleviating the tax burdens on small and medium-sized enterprises. Small and medium-sized enterprises have been the backbone of the Taiwanese economy, accounting for 98% of all registered businesses, 30% of total sales, and 78% of total employment in 2010.⁶ The Income Tax Act Amendment, proposed by the Ministry of Finance, was approved by the cabinet on March 5, 2009, and officially enacted in 2010.⁷

The reform introduced a reduction in marginal statutory tax rates and an increase in the exemption threshold, as illustrated in Figure 1. Before 2010, the corporate income tax schedule consisted of three tax brackets, with a lower threshold of 50 thousand NTD (approximately 1,600 USD). Firms earning below this threshold were exempt from taxation. However, once taxable income surpassed 50 thousand NTD, a statutory tax rate of 15% applied, increasing further to 25% beyond 100 thousand NTD (approximately 40 thousand USD).⁸ The 2010 reform simplified the tax structure by reducing the number of tax brackets from three to two and raising the tax threshold to 120 thousand NTD (approximately 3,900 USD).⁹ Firms with taxable income below this threshold remained exempt from corporate

⁵VAT has exemption rules as well. Specifically, firms with monthly sales below 80 thousand NTD are exempt from VAT and are not included in the administrative dataset. Their sales are instead estimated by the tax authority and subject to a 1% tax rate. Firms with monthly sales between 80 thousand and 200 thousand NTD, certain restaurants, and financial institutions are also exempt from VAT and subject to the aforementioned simplified rules.

⁶The Small and medium-sized enterprises are defined as firms with annual revenues below 80 million NTD (or numbers of employees below 200) in manufacturing, construction, and mining sectors, and those with annual revenues below 100 million NTD (or numbers of employees below 100) in other industries. Source: White Paper on Small and Medium Enterprises in Taiwan accessed at <https://www.sme.gov.tw/article-tw-2853-13128>.

⁷More detailed information can be accessed at <https://www.ey.gov.tw/Page/9277F759E41CCD91/cfe7d369-e943-4a2a-8db8-cedc19384885>.

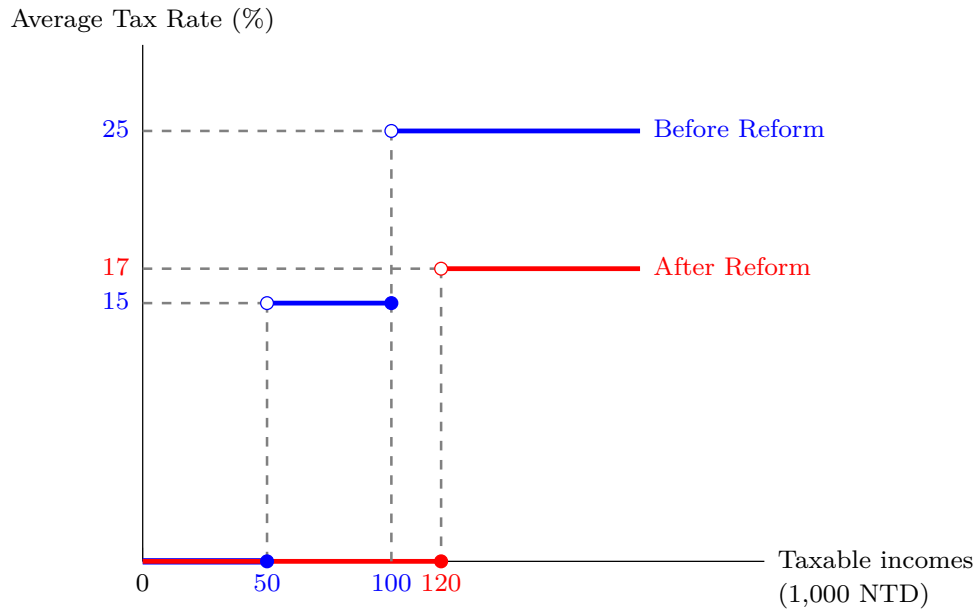
⁸To ease the tax burden on small enterprises, a buffer range was implemented at the first threshold, capping tax liability at 50% of the excess income until the average tax rate reached 15%.

⁹To ensure a gradual transition for firms near the threshold, the government maintained a buffer range, capping corporate tax liability at 50% of the amount exceeding the threshold until the average tax rate

income tax. Above the threshold, a uniform statutory tax rate of 17% applied, replacing the prior tiered structure.

The reform was presented as a pro-growth policy, especially for small and medium-sized enterprises. The new statutory corporate tax rate of 17% was lower than that of mainland China (25%) and South Korea (22%), and closely aligned with Singapore (17%) and Hong Kong (16.5%).¹⁰ However, the reform also sparked public debate. Critics voiced concerns over the potential loss of government revenue, with estimates of annual reductions ranging from 30 billion to 50 billion NTD. They also questioned whether the projected economic gains would be sufficient to offset the fiscal cost of the reform.

Figure 1: Corporate Income Tax Schedule



Notes: This figure shows the corporate income tax schedule in Taiwan. The blue curve denotes the tax schedule before 2010, and the red curve denotes the tax schedule in 2010 and after.

3 Data

Our empirical analysis is primarily based on an employee-employer-linked administrative tax records maintained by the Ministry of Finance in Taiwan from 2004 to 2018. This data set reached 17%.

¹⁰Please see <https://www.lawtw.com/archives/388050>.

covers corporate income tax and value-added tax records, personal income tax records, as well as household registration records. The universal firm coverage, long time series, and rich characteristics allow us to examine how firms respond to corporate income tax reform through both reporting behavior and real production adjustments.

The corporate income tax data provide comprehensive balanced-sheet information, including gross revenues, gross asset (fixed and liquid, including deposit), and a complete list of costs as well as their breakdown items. The value-added tax data provide information on business-to-business transactions and the intermediate input and capital purchases. Moreover, the tax records also cover the basic firm information including ownership, legal entity type, industry, establishment date, and location. Exploiting the panel data structure, we further construct firm revenue growth rate and firm exit. We define firm exit as those disappearing from consecutive tax records. Given the mandatory nature of tax payment, firms that are missing in tax records over consecutive years have likely shut down.

Moreover, the personal income tax data provides comprehensive information on employee salaries. We use firm identifier to link the personal income tax records with the corporate income tax data and the value-added tax data, which yields comprehensive employer-employee linked records. We combine these data sets with household registration records through personal identification numbers, which help identify the kinship relationship between employees and firm owners.

We impose several restrictions to obtain our final sample. First, we keep firms that only subject to standard corporate income tax schedule, but exclude sole proprietorship and partnerships whose profits are subject to the personal income tax at the individual shareholder level.¹¹ Second, we exclude firms with negative taxable incomes, zero reported revenues, or missing employment information, which account for approximately 3% observation. Third, we restrict our sample to firms with taxable incomes between 0 and 300 thousand NTD (around 9,300 USD), covering the major corporate income tax threshold which increased from 50 thousand NTD to 120 thousand NTD.

Table 1 presents summary statistics for firm-level characteristics before and after the 2010 corporate income tax reform. We note four key data patterns. First, despite small taxable incomes, firms in our sample have sizable gross revenue averaged above 15 million NTD.

¹¹We also exclude firms with special filling method such as the Expanded Paper Review and tiny firms that are exempt from the VAT as discussed in Section 2.

Second, intermediate inputs are a dominant component of total cost, accounting for 70–72% of revenue, and labor salaries account for approximately 16% of gross revenue. Third, entry and exit rates are relatively stable across the two periods, with entry rates declining slightly after 2010, suggesting a relative stable yet active market turnover. Fourth, family-run firms are prevalent in our sample, as 70%-80% of firms have shareholders who are immediate family members (spouses, parents, siblings, and children). In addition, family members make up approximately 20% of the workforce, reflecting their significant roles in firm operations.

Table 1: Summary Statistics of Main Sample

	2004-2009		2010-2017	
	Mean (1)	S.D. (2)	Mean (3)	S.D. (4)
<i>Panel A: Revenue and Costs</i>				
Gross revenue (million NTD)	20.0	23.4	16.9	21.3
Intermediate inputs (million NTD)	15.4	20.0	12.6	17.9
Labor salaries (million NTD)	2.5	2.8	2.1	2.6
non-family-member employees	2.0	2.7	1.7	2.4
family-member employees	0.4	0.4	0.4	0.4
Number of employees	11.2	13.3	8.6	11.4
non-family-member employees	9.8	13.2	7.4	11.4
family-member employees	1.4	1.0	1.2	1.0
Capital investment Cost (million NTD)	0.3	2.2	0.4	3.6
Taxable Profit (thousand NTD)	81	76	103	72
<i>Panel B: Firm Dynamics</i>				
Revenue growth rate	0.11	0.67	0.09	0.62
Entry rate	0.10	0.29	0.07	0.10
Exit rate	0.10	0.30	0.11	0.10
Firm age	11.3	8.8	14.8	10.6
Share of family firms	0.73	0.28	0.79	0.27
Number of observations	255,539		332,068	

Notes: This table shows summary statistics before and after the 2010 corporate income tax reform (pre-reform pooled 2004-2009 in columns (1) and (2), and post-reform pooled 2010-2017 in columns (3) and (4).

4 How Firms Respond to Tax Notches

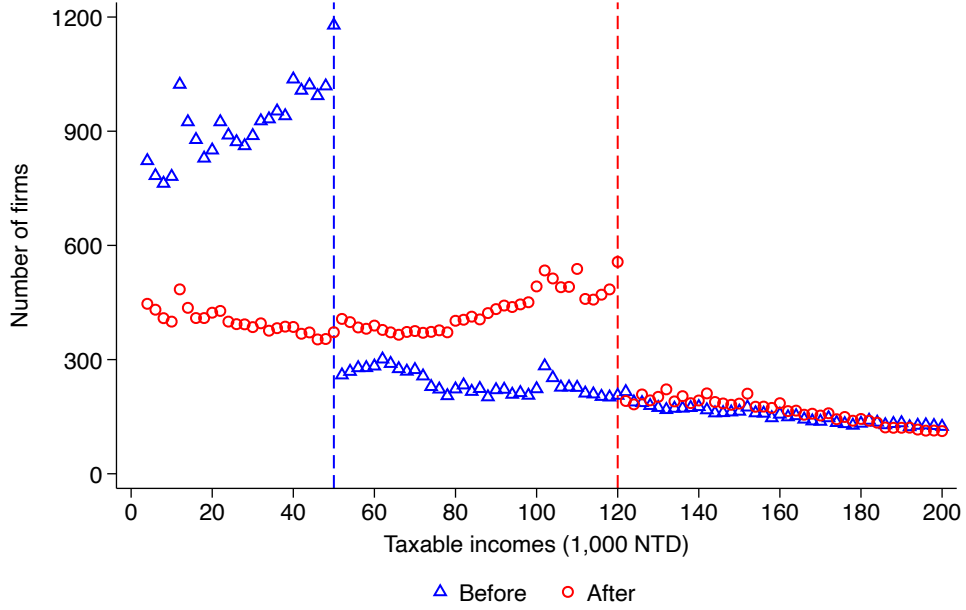
4.1 Bunching below the Exemption Threshold

We begin by examining the distribution of corporate taxable income as shown in Figure 2. The figure plots the number of firms across taxable income bins (2,000 NTD), averaged over the years before the reform (2004–2009, in blue triangles) and after the reform (2010–2016, in red circles). Before the corporate income tax reform, a clear bunching pattern appears at the exemption threshold of 50 thousand NTD. The number of firms increases gradually as taxable income approaches the threshold and then drops sharply above it, from roughly 900 firms in each 2,000-NTD taxable income bin below the threshold to fewer than 300 just above. Since the pre-reform tax schedule had been unchanged for over twenty years, the discontinuity in the distribution of corporate taxable incomes indicates strong and persistent incentives for firms to strategically manage their corporate taxable incomes to remain just below the exemption threshold and thereby avoid corporate income taxes.

More importantly, the distribution of taxable income shifts noticeably in response to the change in the exemption threshold, indicating that firms adjusted their reported income around notches in the corporate tax schedule. After the reform, the discontinuity at the old threshold of 50 thousand NTD disappears entirely, while a new concentration of firms emerges just below the new exemption threshold of 120 thousand NTD. On average, there are approximately 420 firms in each 2,000-NTD taxable income bin below the new exemption threshold, with a modest uptick in counts in the bins closer to this threshold. However, there are approximately 165 firms per bin immediately above the exemption threshold. Among firms with taxable incomes below 120 thousand NTD, the share of firms below 50 thousand NTD decreased substantially from 73% to 39%. Those firms mainly increased their taxable incomes above 50 thousand NTD, with the share of firms reporting taxable incomes between 50 thousand NTD and 120 NTD increasing from 27% to 61%. In Appendix B, we quantify firm responses to the tax notches following the bunching literature (Saez, 2010; Kleven and Waseem, 2013; Almunia and Lopez-Rodriguez, 2018).

Beyond the pronounced bunching patterns at the exemption thresholds, a number of other noteworthy empirical patterns arise. First, we observe diffuse bunching with a noticeable concentration of firms just below the exemption threshold accompanied by a non-zero mass just above it, rather than a sharp spike. This pattern is consistent with the literature

Figure 2: Distribution of Corporate Taxable Incomes



Notes: This figure shows the average annual distribution of corporate taxable incomes before the reform (2004-2009, in blue triangles) and after (2010-2017, in red circles). The vertical axis denotes the average number of firms yearly in each taxable income bin. The horizontal axis represents the taxable income bins. The bins are \$20,000 wide, delimited such that no bin contains data both to the left and to the right of the relevant policy thresholds. For example, the bin 1 represents firms that have taxable income between 10 thousand NTD (included) and 12 thousand NTD (not included).

on bunching under frictions such as search costs or lumpy adjustment ([Anagol, Davids, Lockwood, and Ramadorai, 2024](#)).

Second, firms primarily respond to the exemption threshold but show little adjustment around the higher threshold in the tax schedule. For example, Figure 1 shows a tax notch at 100 thousand NTD prior to the reform where the average tax rate increased from 15% to 25%, but we find little evidence of bunching at this higher threshold. This limited response is consistent with the literature ([Kleven and Waseem, 2013](#)) and can be attributed to tax-rate misperception and other optimization frictions.

Third, firms' bunching responses are immediate and common across industries. The distribution of taxable income changed sharply between 2009 and 2010, as shown in Appendix Figure A.2, and remained roughly stable since 2010. The sudden increase in the number of firms reporting taxable incomes between 50 thousand and 120 thousand NTD in

2010 are mainly from those reporting between 30 thousand and 50 thousand NTD in 2009, according to Panel (a) of Appendix Figure A.3. Moreover, the bunching pattern remains consistent across sectors, since we observe similar bunching behaviors in wholesale, retail, manufacturing, and construction sectors, as shown in Appendix Figure A.4.

Finally, although the reform also reduced the statutory tax rate from 25% to 15% for firms with taxable incomes above 120 thousand NTD, the number of firms reporting taxable incomes between 120 thousand and 200 thousand NTD remains largely unchanged. Therefore, our analysis focuses on firms' behavioral responses to the exemption threshold and the dynamic implications of these responses. Firm responses to marginal tax rate changes above 120 thousand NTD are interesting in their own right (similar to evaluations on the Tax Cuts and Jobs Act in the US (Kennedy, Dobridge, Landefeld, and Mortenson, 2022)), but we leave the detailed empirical investigation to future work.

4.2 How Do Firms Bunch?

We next examine how firms adjust their taxable incomes to bunch below the exemption threshold. We demonstrate two primary channels underlying the bunching behavior: real responses by shrinking production size, and misreporting by shifting profits as labor expenditures. We distinguish these two channels by examining average shares of intermediate input, investment, and labor salary, relative to gross revenues. Figure 3 shows the average input cost shares in gross revenues with respect to the reported corporate incomes before and after the reform, respectively.

Real responses Panel (a) of Figure 3 plots the ratio of intermediate inputs and revenues across taxable income bins. We observe that firms with taxable incomes just below the exemption threshold have significantly smaller intermediate input per unit of gross revenues, both before and after the reform. Since the discontinuities in the ratios of intermediate inputs and revenues locate exactly at the old exemption threshold before the reform, and moves sharply with the shift of the new exemption threshold after the reform, we attribute the discontinuity to the corporate income tax notches and its changes. Other concurrent policies do not coincide with the corporate income tax reforms either in the notched structure or in the timing. Therefore, this pattern suggests that firms below the exemption threshold tend to scale down production by utilizing less intermediate input. Moreover, we plot the ratio

of investment expenditures and revenues across taxable income bins in Panel (c) of Figure 3. It shows that the total investment exhibits a similar pattern, albeit less pronounced due to larger adjustment cost for capital.

Profit shifting We plot the ratio of labor salaries and revenues across taxable income bins in Panel (b) of Figure 3. In contrast to the data patterns of intermediate inputs and capital investment, we find that firms with taxable incomes just below the exemption threshold have significantly higher labor salary per unit of gross revenues. This stark discontinuity suggests that firms strategically over-report labor costs to lower taxable income. We further examine which group of employees enjoy an increase in the salary, by exploiting rich information in our employee-employer linked records with family registration. We classify employees into two groups: those related to the firm’s shareholder (family members) and those not related (non-family members).¹² As shown in Appendix Figure A.5, we find that firms mainly shift corporate profits as personal incomes paid to relatives of firm owners. Since the personal income tax (5%) is much lower than the corporate income tax above the exemption threshold (15% or 25% before the reform, 17% after the reform), the profit shifting behavior creates lucrative opportunities for firms to avoid paying corporate income taxes and lower the overall tax burdens.

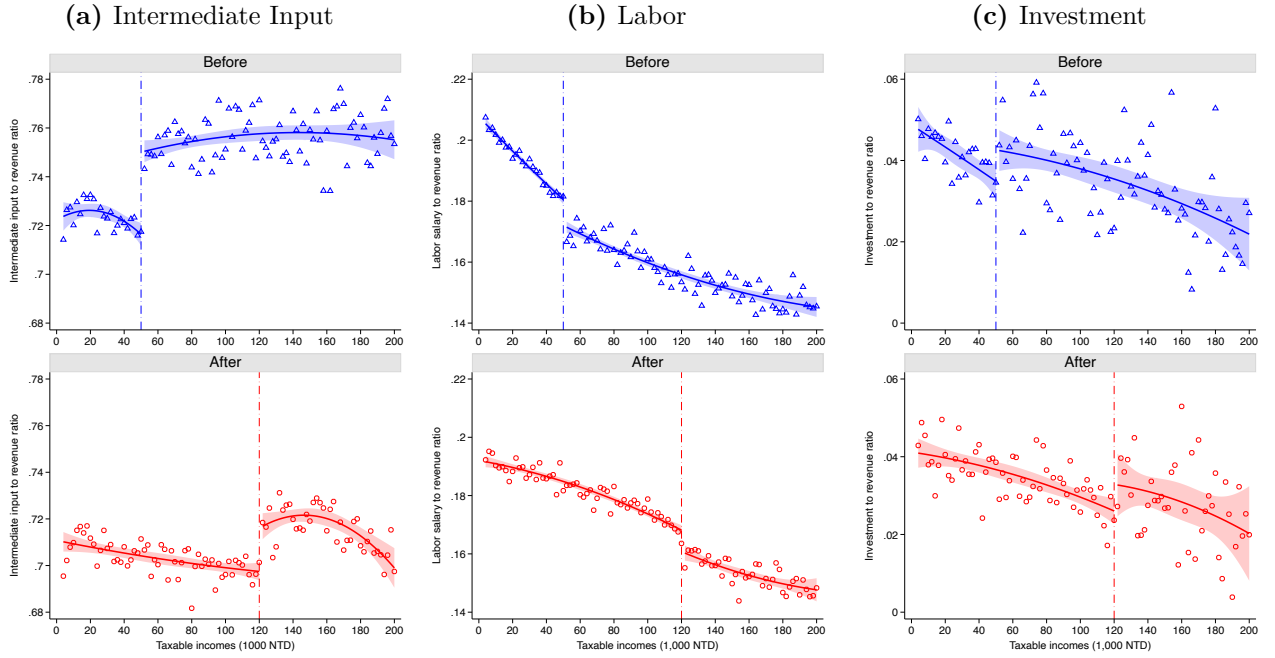
Other channels We further investigate potential alternative mechanisms and find no evidence that firms strategically lower their reported taxable income by manipulating revenues. Average firm revenues across taxable-income bins increase smoothly with taxable incomes and show no visible discontinuity around the thresholds, as illustrated Panel (a) in Appendix Figure A.6. If firms were managing their taxable incomes primarily by adjusting reported revenues, we would expect to observe a sharp decline in total revenues just below the threshold. However, the data reveals no such pattern, suggesting that revenue manipulation is unlikely to be the main channel driving firms’ bunching behavior.

To further address concerns about revenue under-reporting, we decompose firms’ total revenue into business-to-business (B2B) and business-to-consumer (B2C) components and construct the ratio of B2B revenues to total revenues. Under the value-added tax system, if firms under-report revenues, they would likely conceal B2C sales while leaving B2B transactions largely unchanged.¹³ As shown in Panel (b) of Appendix Figure A.6, the B2B revenue

¹²We define relatives as the spouse, parents, siblings, or kids of the focal firm owner.

¹³The value-added tax system functions as a third-party reporting mechanism that enhances traceability

Figure 3: Input Revenue Ratios and Taxable Income



Notes: This figure shows the average input revenue ratios with respect to corporate taxable income bins before the reform (2004–2009, upper panels with blue triangles and lines) and after (2010–2017, lower panels with red circles and lines). Panel (a) plots the ratios of intermediate input and gross revenue; Panel (b) plots the ratios of labor salary and gross revenue; Panel (c) plots the ratios of investment and gross revenue. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

share remains consistently around 70% across all taxable income bins, both before and after the reform, providing no evidence of systematic under-reporting through this channel.

Finally, we rule out the possibility that firms reduce their taxable income by splitting into multiple smaller entities. Using shareholder information, we identify “splitting” firms as those sharing at least one common shareholder with another firm and calculate the ratio of such firms to the total number of firms in each taxable-income bin. As shown in Appendix Figure A.7, this ratio remains stable at around 10–15% across all bins and over time, with no noticeable discontinuity at the threshold or change following the reform. These results

for B2B transactions. Because B2B sales represent deductible inputs for buyer firms, they have an incentive to report these transactions to the tax authority. In absence of collusion between supplier and buyer firms, discrepancies in B2B sales are easily cross-checked, making them less prone to under-reporting. In contrast, B2C transactions are much less traceable, as final consumers lack incentives to retain or report receipts. See Kleven, Knudsen, Kreiner, Pedersen, and Saez (2011) and Carrillo, Pomeranz, and Singhal (2017) for discussions of transaction traceability and evasion.

suggest that ownership restructuring or firm splitting is unlikely to be the main driver of the empirical patterns we document.

4.3 Dynamic Consequences: Growth and Exit

We next examine the dynamic consequences of the bunching behavior. By comparing firms with taxable income immediately below and above the exemption thresholds, we document two related dynamic effects for those bunching firms. First, they tend to have a lower revenue growth rate. Second, they tend to have a higher exit rate.

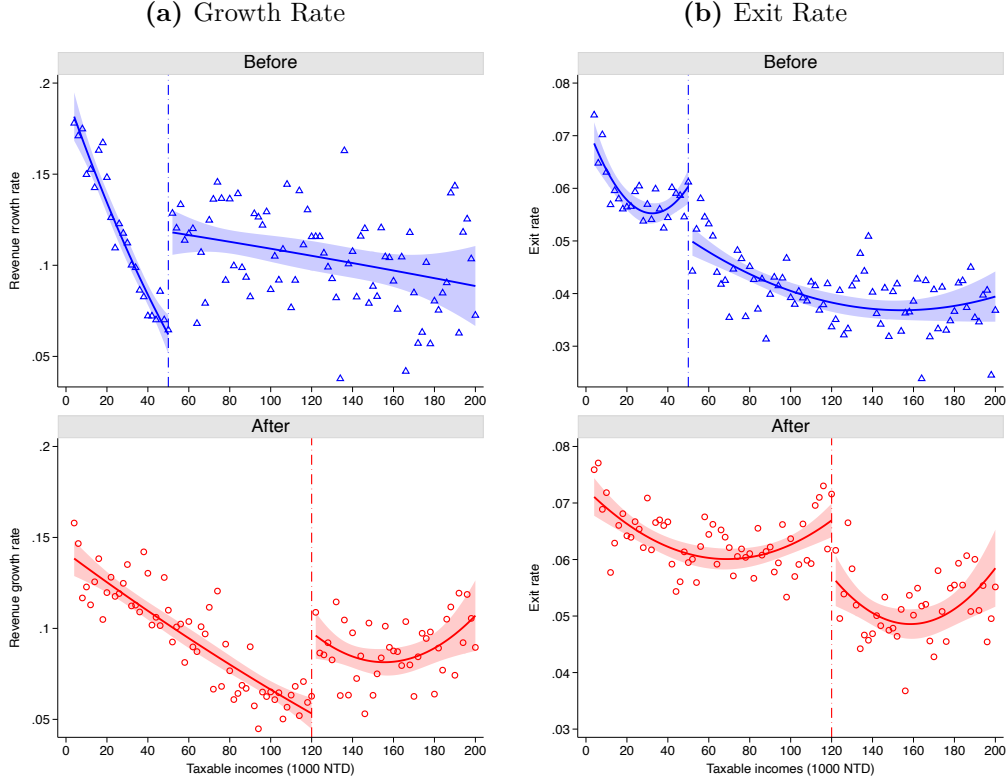
Slow-down of revenue growth Panel (a) of Figure 4 plots the annual revenue growth rate with respect to taxable income bins. We define a firm's revenue growth rate as the percentage changes in gross revenues between two adjacent years. We find that firms experience a slowdown in revenue growth if their taxable income is below the exemption threshold, and there is a sharp increase in the revenue growth rate for firms with taxable incomes above the exemption threshold. Similar to static decision, the discontinuity in the revenue growth rate moves sharply to the shift of the exemption threshold after the reform. We substantiate this pattern as a real response rather than a reporting behavior by examine the growth rate of revenues from business-to-business transactions and intermediate inputs, both of which are under scrutiny of the value-added tax system. As shown in Appendix Figure A.8, the pattern is similar to what we observe in Figure 4.

Increased exit rate Panel (b) of Figure 4 shows the average exit rate across taxable income bins. We define firm exit as stopping submitting tax records. We find that firms whose taxable income is below the exemption threshold have a significantly higher exit rate, and there is a sharp drop in the exit rate for firms with taxable income above the exemption threshold. Using alternative definitions of firm exit, such as missing tax record in a certain year, or for three consecutive years, we find that the data patterns in Appendix Figure A.9 are similar to our baseline pattern.

Firms which bunch below the exemption thresholds have shrunk their operation to be lower than the optimal level in the absence of tax incentives. These bunching firms require extra favorable growth opportunities to justify growing beyond the exemption thresholds and incurring substantially higher tax liabilities. Therefore, they tend to remain small in size, but will become more vulnerable to adverse cost shocks and are more likely to be eliminated

from the market. We thus uncover evidence of a “bunching trap” created by tax notches, where firms become persistently stagnant and eventually exit.

Figure 4: Revenue Growth Rate, Exit Rate, and Taxable Income



Notes: This figure shows the average revenue growth rates and exit rates with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the average revenue growth rates; Panel (b) plots the exit rate where firm exit is defined as disappearing from the tax records. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

4.4 Regression Analysis Based on the 2010 Tax Reform

Empirical strategy Our descriptive evidence rely on a cross-sectional variation induced by the exemption thresholds before and after the 2010 corporate income tax reform, respectively. We argue that our descriptive evidence documents the impacts of the tax notches, since we observe sharp discontinuities exactly at the exemption thresholds, which is unlikely

to be an artifact from any other policy changes. To further boost the confidence in our empirical findings, we complement bunching-based analyses with a regression framework, which relies on an exogenous shift in the *relative distance* of firms' taxable incomes and the exemption threshold due to the 2010 corporate income tax reform. We include a rich set of fixed effects in the regression model to control for other potential confounding factors. Our empirical model is as below.

$$y_{ijt} = \sum_{k=-5, k \neq 1}^5 \alpha_k \mathbb{1}(BinDistance_{it} = k) + \sum_{b=1}^{20} \beta_b \mathbb{1}(Bin_{it} = b) + \lambda_{jt} + X_{it} + \epsilon_{ijt} \quad (1)$$

We denote the outcome variable as y_{ijt} , including the average input cost shares in gross revenues, revenue growth rate, or the exit dummy for firm i in industry j and year t . We group firms into taxable income bins of 10 thousand NTD, Bin_{it} . For example, the group of firms with $Bin_{it} = 4$ include those with taxable incomes between 40 thousand (included) and 50 thousand NTD (not included). The variable $BinDistance_{it}$ measures the gap between the exemption threshold and firm i 's taxable income such that

$$BinDistance_{it} = \begin{cases} 5 - Bin_{it}, & \text{if } t \leq 2009 \\ 12 - Bin_{it}, & \text{if } t > 2009. \end{cases}$$

A negative value of $BinDistance_{it}$ indicates that the taxable income of firm i is above the exemption threshold. For example, $BinDistance_{it} = -2$ for firms with $Bin_{it} = 7$ (taxable income between 70 thousand and 80 thousand NTD) in 2009, since the exemption threshold was 50 thousand NTD. $BinDistance_{it} = 5$ for firms with $Bin_{it} = 7$ (taxable income between 70 thousand and 80 thousand NTD) in 2010, because the exemption threshold increased to 120 thousand NTD.

We include a full set of dummies of $BinDistance_{it}$ and Bin_{it} . Coefficients α_k are of key interests as they capture how tax notches shape firm outcomes. The key identification assumption is that conditional on the firms' taxable income bins Bin_{it} , the variation in the relative distance to the exemption threshold $BinDistance_{it}$ comes from the exogenous shift in the threshold from 50 thousand NTD to 120 thousand NTD by the 2010 corporate income tax reform. Consequently, α_k compares how firm outcomes change with the distance to the

exemption threshold, even holding their taxable income bins constant.

Moreover, we control for industry-by-year fixed effects, λ_{jt} , to account for common macroeconomic shocks over time and across industries, such as economic stimulus policies, trade shocks, inflation, or labor market conditions that could affect compensation for all firms. These fixed effects control for the impact of the 2008 global financial crisis and the subsequent economic recovery as well. The vector X_{it} includes firm-level covariates, including a firm’s location, company type, age, and lagged assets, which capture potential differences in outcomes driven by firm characteristics unrelated to the relative distance between a firm’s taxable income and the exemption threshold.

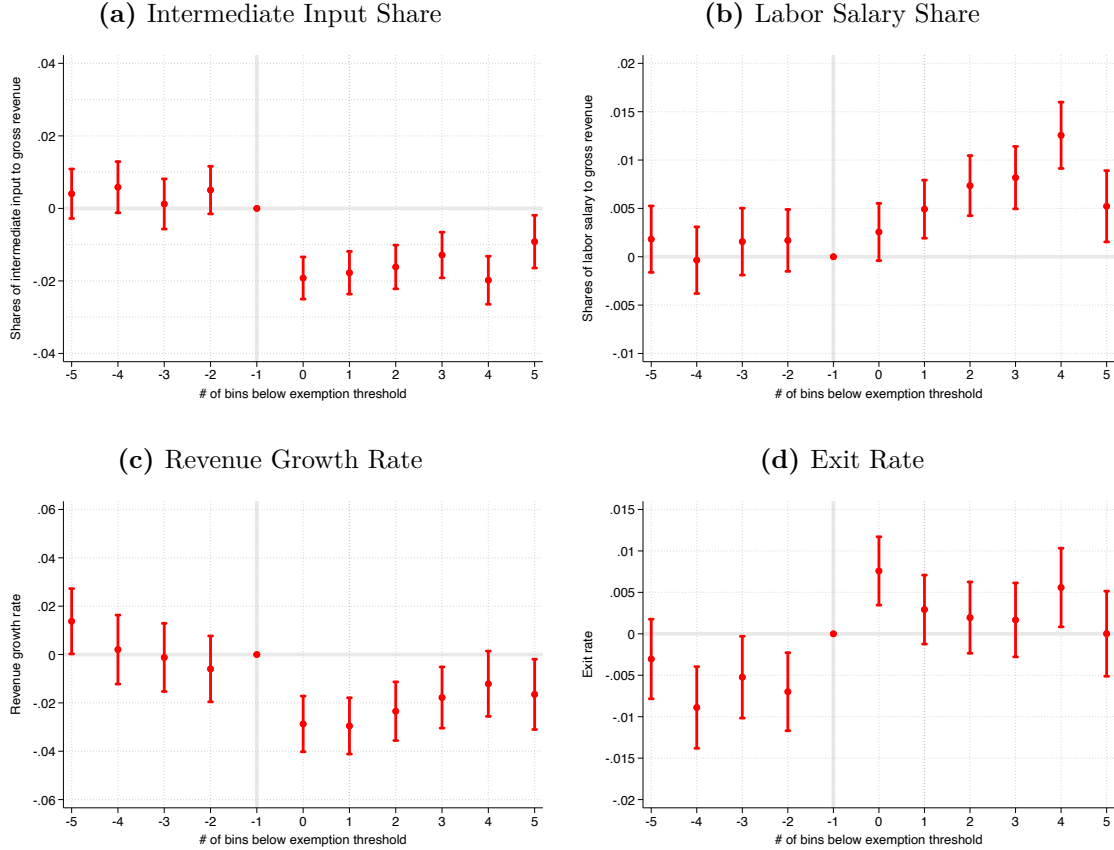
Estimation results We present results in Figure 5, which are consistent with the empirical patterns in Figures 3 and 4. Compared to firms above the exemption threshold ($BinDistance_{it} < 0$), firms whose taxable incomes are below the exemption threshold tend to have a significantly lower intermediate input share in gross revenue, conditional on firm size, industry-specific macro shocks, and other firm-level confounding factors. The average intermediate input share in gross revenues is approximately 2% ($= \frac{0.015}{0.75}$) lower for firms that are below the exemption threshold, which is sizable considering that the average taxable income to gross revenue ratio is 0.4%. Moreover, compared to firms above the exemption threshold ($BinDistance_{it} < 0$), firms whose taxable incomes are below the exemption threshold tend to have a significantly higher labor salary share in gross revenue. The average labor salary share in gross revenues is approximately 6.7% ($= \frac{0.01}{0.13}$) higher for firms that are below the exemption threshold. These results confirm the existence of both real responses and profit shifting for firms to avoid corporate income taxes.

Panels (c) and (d) further demonstrate the dynamic consequences of bunching due to tax notches. We find that firms whose taxable incomes are below the exemption threshold tend to have a much lower revenue growth rate compared with their counterparts above the threshold. The revenue growth rate is 2.1 percentage points lower, which is approximately 20% of the sample mean. Moreover, firms below the exemption threshold also have a higher exit rate, which is 0.7 percentage points higher and approximately 8.1% of the sample mean.

The baseline empirical design relies on variation between firms within a firm size class. As a robustness check, we implement an alternative strategy that exploits within-firm variation in the same spirit of the traditional difference-in-differences design. We defer a detailed description to Appendix C. We examine the effects of lifting the exemption threshold from the

2010 reform on firms' intermediate input share and labor compensation shares, controlling for firm fixed effects. The results are fully consistent with our main findings. The same empirical framework is not well suited to study firms' dynamic responses, as we cannot control for firm fixed effects when analyzing exit decisions.

Figure 5: Empirical Evidence Exploiting Changes in Distance to Threshold



Notes: The figure displays the coefficient estimates for α_k in equation (1) with 95% confidence intervals, capturing the treatment effect of tax notches on firm outcomes. The empirical model controls for firm corporate income bin dummies, industry-year fixed effects, and firm-level covariates, including location, company type, age, and lagged assets. The horizontal axis represents the relative distance of firms' taxable income from the exemption threshold, and positive number represents taxable income bins below the threshold. For example, the group with the relative distance 0 includes firms with taxable income between 40 thousand and 50 thousand NTD before the reform (when the exemption threshold was 50 thousand NTD) and firms with taxable income in the range of 110 thousand and 120 thousand NTD after the reform (when the exemption threshold was raised to 120 thousand NTD).

5 A Structural Model of Firm Production, Reporting, and Exit

In this section, we build an industry dynamics model incorporating firms' decision on static production and tax reporting, as well as dynamic capital adjustment and exit. At the beginning of each year, firms are endowed with heterogeneous capital and productivity. They earn static profits but can potentially avoid corporate income taxes by changing firm production or misreporting labor salary. They will then decide whether to exit, or how much to invest if they decide to stay in operation. We first introduce the static decision of firm production and tax reporting and then describe the firms' dynamic decision as well as the industrial equilibrium.

A firm is index by j . Time is discrete and a period is a year denoted by t . Our model focuses on small and micro enterprises and assumes firms operate in both competitive output and input markets, the prices of which are determined exogenously by aggregate demand and supply.

5.1 Production and Reporting

Each firm j has a realized productivity z_{jt} at the beginning of the year t . Conditional on the realized productivity z_{jt} and capital stock k_{jt} , firms make the optimal variable input choices in face with competitive input prices, and decide how much output to produce on the competitive output market. We assume firms produce output y_{jt} with the capital k_{jt} , labor l_{jt} , and intermediate input m_{jt} , according to the following production function.

$$y_{jt} = z_{jt} k_{jt}^{\alpha_k} l_{jt}^{\alpha_l} m_{jt}^{\alpha_m}.$$

We use α_k , α_l , and α_m as input intensity parameters for capital, labor, and intermediate inputs, respectively. We denote $\lambda = \frac{1}{\alpha_m + \alpha_l} > 1$, and input price aggregator as Φ_t .¹⁴ The

¹⁴If we denote the intermediate input price as p_t and wage rate as w_t , we could express the price aggregator Φ_t as follows.

$$\Phi_t = \frac{p_t^{\alpha_m \lambda} w_t^{\alpha_l \lambda}}{\lambda \alpha_m^{\alpha_m \lambda} \alpha_l^{\alpha_l \lambda}}.$$

input cost function is the follows.

$$c(y_{jt}, z_{jt}, k_{jt}) = \Phi_t \left(\frac{y_{jt}}{z_{jt} k_{jt}^{\alpha_k}} \right)^\lambda$$

Specifically, the optimal labor cost $c^l(y_{jt}, k_{jt}, z_{jt})$ and the optimal intermediate input cost $c^m(y_{jt}, k_{jt}, z_{jt})$ can be expressed as follows.

$$c^l(y_{jt}, k_{jt}, z_{jt}) = \alpha_l \lambda c(y_{jt}, z_{jt}, k_{jt}), \quad c^m(y_{jt}, k_{jt}, z_{jt}) = \alpha_m \lambda c(y_{jt}, z_{jt}, k_{jt}).$$

We normalize the output price as 1, and thus firm profit function is defined as the follows.

$$\pi_{jt} = \pi(y_{jt}, k_{jt}, z_{jt}) = y_{jt} - c(y_{jt}, k_{jt}, z_{jt}).$$

The optimized output y_{jt}^* and firm optimal profit π^* satisfy the following two conditions.

$$y_{jt}^* = \left[\frac{(z_{jt} k_{jt}^{\alpha_k})^\lambda}{\lambda \Phi_t} \right]^{\frac{1}{\lambda-1}}, \quad \pi_{jt}^* = (\lambda - 1) \Phi_t \left(\frac{y_{jt}}{z_{jt} k_{jt}^{\alpha_k}} \right)^\lambda.$$

The government introduces corporate income tax schedule with a notch such that firms with profit π_{jt} below $\bar{\pi}$ would enjoy a low tax rate τ_L , while firms with profit π_{jt} above $\bar{\pi}$ are subject to a high tax rate τ_H , and $\tau_L < \tau_H$.

$$\tau_{jt} = \begin{cases} \tau_L, & \text{if } \pi_{jt} \leq \bar{\pi} \\ \tau_H, & \text{if } \pi_{jt} > \bar{\pi} \end{cases}$$

Therefore, we wrote $\tau_{jt} = \tau_L \times \mathbb{1}(\pi_{jt} \leq \bar{\pi}) + \tau_H \times \mathbb{1}(\pi_{jt} > \bar{\pi})$. Each period, firms decide whether to bunch below $\bar{\pi}$ and enjoy the low tax rate τ_L . Firms have two strategies to achieve bunching. First, firms could shrink their production size by choosing y_{jt} lower than y_{jt}^* . Second, firms could shift their profit to labor salary paid to their relatives and choose a *reported* labor expenditure c_{jt}^r higher than optimal labor expenditure $c_{jt}^l = c^l(y_{jt}, k_{jt}, z_{jt})$. We assume that firms pay the profit shifting cost as follows.

$$g(c_{jt}^r, c_{jt}^l, \epsilon_{jt}) = \frac{\lambda(c_{jt}^r - c_{jt}^l)^2}{c_{jt}^l} + \theta_\epsilon \epsilon_{jt}.$$

λ captures the magnitude of misreporting costs. ϵ_{jt} is an i.i.d. component of the fixed profit shifting cost, which follows an exponential distribution with mean θ_ϵ .

Firms trade off the potential savings on the tax payment against the profit-shifting cost and the distortion on production. We denote the bunching decision as d_{jt} which equals one if the firm chooses to bunch the profit below $\bar{\pi}$. We allow firms to “over-bunch” since the data reveals that the final reported profit under bunching will not be precisely at $\bar{\pi}$ because of reporting uncertainty or to avoid scrutiny. Instead, firms will choose the final reported profit as $\bar{\pi}_{jt} = \bar{\pi} - \omega_{jt}$, where ω_{jt} follows a i.i.d bounded exponential distribution with mean θ_ω . Firm’s profit maximization problem subject to the tax notch can be formulated as below.

$$\begin{aligned} \Pi_{jt} = & \max_{y_{jt}, c_{jt}^r, d_{jt} \in \{0,1\}} (1 - d_{jt}) \times (1 - \tau_{jt}) \times \pi(y_{jt}^*) + d_{jt} \times (1 - \tau_L) \times [\pi(y_{jt}) - g(c_{jt}^r, c^l(y_{jt}), \epsilon_{jt})] \\ \text{s.t. } & \pi(y_{jt}) + c^l(y_{jt}) - c_{jt}^r = \bar{\pi}_{jt} \leq \bar{\pi}. \end{aligned}$$

Therefore, the ratio of firms that decide to bunch is given by the follows, where $\tilde{y}_{jt}$ and $\tilde{c}_{jt}^r$ denote the optimal production and reported labor salary if a firm decides to bunch.

$$P(d_{jt} = 1) = 1 - \exp\left\{-\frac{\pi(\tilde{y}_{jt}) - \frac{\lambda[\tilde{c}_{jt}^r - c^l(\tilde{y}_{jt})]^2}{c^l(\tilde{y}_{jt})} - \frac{(1-\tau_{jt})\pi(y_{jt}^*)}{1-\tau_L}}{\theta_\epsilon}\right\} \quad (2)$$

The reported profit π_{jt}^r and real revenue y_{jt}^r , given the optimal bunching choice $\tilde{d}_{jt}$, is as follows.

$$\pi_{jt}^r = (1 - \tilde{d}_{jt}) \times \pi(y_{jt}^*) + \tilde{d}_{jt} \times \pi(\tilde{y}_{jt}); \quad y_{jt}^r = (1 - \tilde{d}_{jt}) \times y_{jt}^* + \tilde{d}_{jt} \times \tilde{y}_{jt}.$$

The reported intermediate input and labor salary are as below.

$$c_{jt}^{mr} = (1 - \tilde{d}_{jt}) \times c^m(y_{jt}^*) + \tilde{d}_{jt} \times c^m(\tilde{y}_{jt}); \quad c_{jt}^{lr} = (1 - \tilde{d}_{jt}) \times c^l(y_{jt}^*) + \tilde{d}_{jt} \times c^l(\tilde{y}_{jt}).$$

The notches in the corporate income tax schedule introduce non-linear incentives for them to bunch below the tax threshold, steer their reporting behavior, and distort their production choices. We denote the real profit conditional on the random draw of ϵ_{jt} and ω_{jt} as $\Pi(k_{jt}, z_{jt}, \epsilon_{jt}, \omega_{jt})$. The optimized ex-ante static profit is thus $\Pi(k_{jt}, z_{jt}) = \mathbb{E}_{\epsilon, \omega}[\Pi(k_{jt}, z_{jt}, \epsilon_{jt}, \omega_{jt})]$.

5.2 Growth, Exit, and Entry

Incumbent firms decide capital investment and exit after earning static profit. Each firm incurs a scrape value ξ_{jt} from a common distribution $G(\xi_{jt})$ and decides whether to exit the market by comparing the continuation value with the scrape value. If the continuation value falls below the scrape value, it will exit the market and earn ξ_{jt} . Otherwise, the firm will stay on the market and decide how much to invest i_{jt} , subject to convex investment costs $C^i(i_{jt}, k_{jt})$. The capital stock in the next period k_{jt+1} evolves according to the current capital stock and the investment, which in turn affects the optimal output in the next period as well as firm revenue growth rate. Firm productivity follows a Markov transition process following the conditional distribution $F(z_{jt}|z_{jt-1})$.

The value function is as follows.

$$V(k_{jt}, z_{jt}, \xi_{jt}) = \Pi(k_{jt}, z_{jt}) + \max\{\xi_{jt}, \max_{i_{jt}} \mathbb{E}_\nu[-C^i(i_{jt}, k_{jt}, \nu_{jt}) + \beta \mathbb{E}(V(k_{jt+1}, z_{jt+1})|k_{jt}, z_{jt})]\}$$

$$\text{s.t. } k_{jt+1} = \delta k_{jt} + i_{jt}$$

We parameterize the capital adjustment cost as a quadratic function with respect to the investment i_{jt} where ϕ captures the marginal capital investment cost, $C^i(i_{jt}, k_{jt}, \nu_{jt}) = \frac{\phi_2 i_{jt}^2}{k_{jt}} + \phi_1 i_{jt} + \nu_{jt} i_{jt}$. Therefore, the optimal investment i_{jt}^* follows the condition below.

$$\frac{\partial \mathbb{E}_\nu C^i(i_{jt}^*, k_{jt}, \nu_{jt})}{\partial i_{jt}} \leq \frac{\partial \beta \mathbb{E}(V(\delta k_{jt} + i_{jt}^*, z_{jt+1})|k_{jt}, z_{jt})}{\partial i_{jt}}$$

The equality holds if the optimal investment i_{jt}^* is positive. Moreover, the exit rate $P(X_{jt} = 1)$ with X_{jt} defined as the exit decision, can be formulated as follows.

$$P(X_{jt} = 1) = 1 - \exp\left\{-\frac{\max_{i_{jt}} \mathbb{E}_\nu[-C^i(i_{jt}, k_{jt}, \nu_{jt}) + \beta \mathbb{E}(V(k_{jt+1}, z_{jt+1})|k_{jt}, z_{jt})]}{\theta_\xi}\right\} \quad (3)$$

The revenue growth rate is defined as follows.

$$\% \Delta \tilde{y}_{jt} = \frac{\mathbb{E}(\tilde{y}_{jt+1}) - \tilde{y}_{jt}}{\tilde{y}_{jt}} = \frac{\mathbb{E}[\tilde{y}(k_{jt+1}^*, z_{jt+1})|k_{jt}, z_{jt}] - \tilde{y}_{jt}}{\tilde{y}_{jt}}.$$

6 Next Steps and Conclusion

The estimation of the structural model is still in progress. Our goal is to use the structural model to quantify the welfare consequences from the tax notches and evaluate winners and losers from the tax reform. Moreover, we aim to explore alternative designs of the tax schedule for various policy goals, trading off between tax revenues and alleviating the tax burdens especially for small and medium enterprises.

In this paper, we examine firms' responses and the aggregate policy implications of notched corporate income tax design, using evidence from a corporate income tax reform in Taiwan. Empirically, we find that firms respond strongly to changes in the exemption threshold on the corporate income tax schedule, employing a mix of real and reporting strategies. On one hand, firms over-report salaries for employees who are family members of the firm owner, inflating deductions to lower taxable income. On the other hand, firms reduce production by strategically and slow down their revenue growth. The stagnant growth further leads to an unexpected increase in exit probability.

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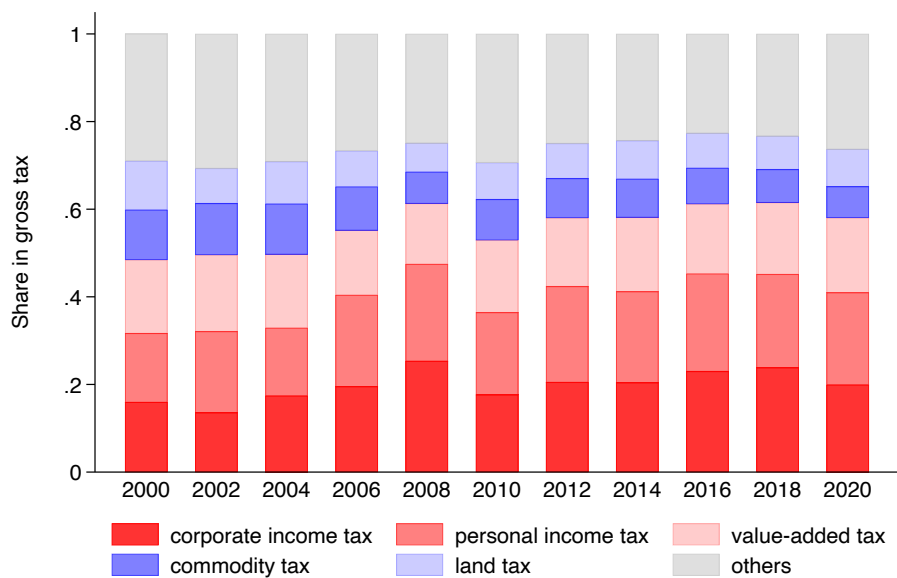
Supplemental Appendix for Notch and Catch: Dynamic Inefficiency in the Corporate Income Tax Schedule

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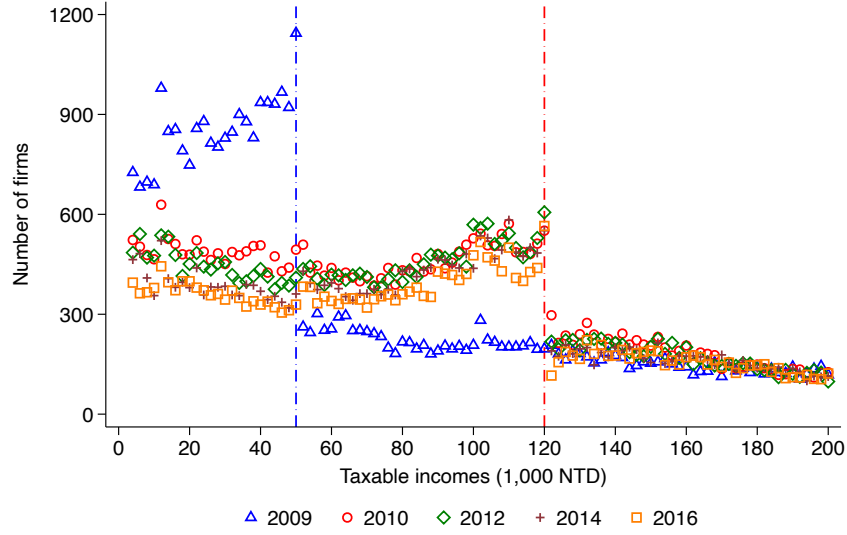
A Additional Figures

Figure A.1: Different Sources of Gross Tax Revenue in Taiwan 2000-2020



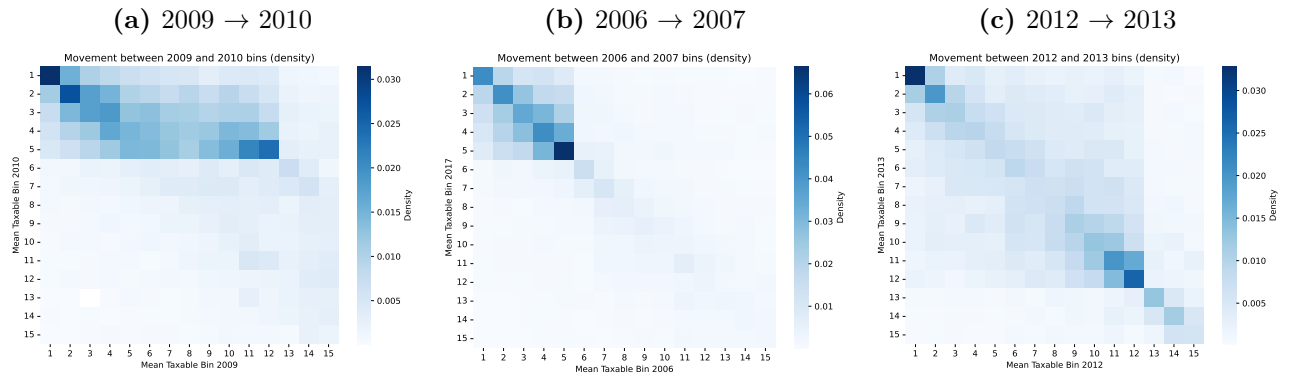
Notes: This figure plots the shares of different types of taxes to gross tax revenues in Taiwan between 2000 and 2020. The data source is the Public Finance Statistics Database in Taiwan accessed at <https://web02.mof.gov.tw/njswww/webmain.aspx?sys=100&funid=edefjspf2>.

Figure A.2: Distribution of Corporate Taxable Incomes by Years



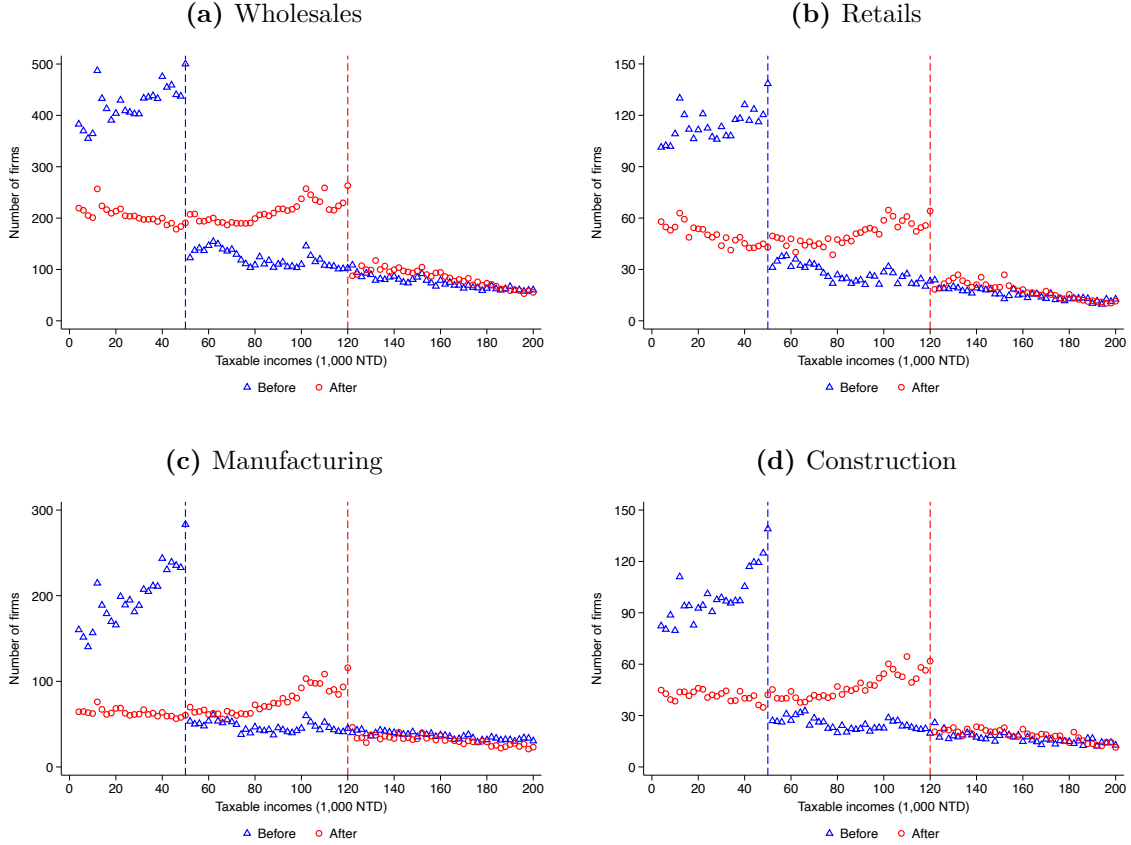
Notes: This figure shows the average annual distribution of corporate taxable incomes across years between 2009 and 2016. The vertical axis denotes the average number of firms yearly in each taxable income bin. The horizontal axis represents the taxable income bins.

Figure A.3: Density of Firms Moving across Taxable Income Bins



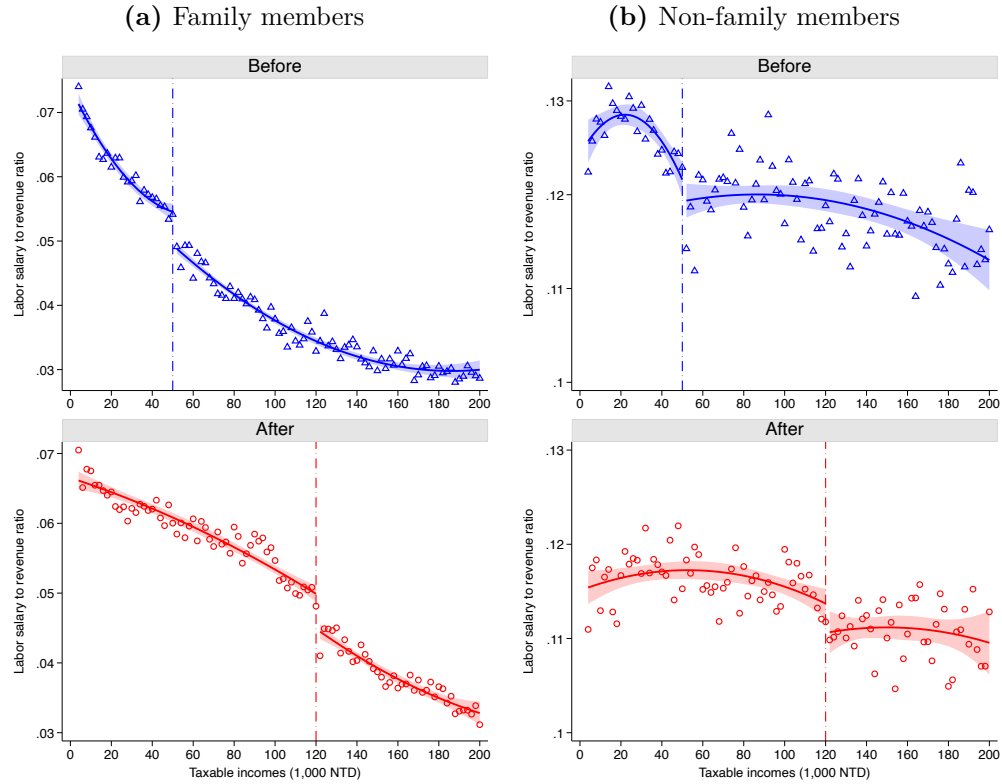
Notes: This figure presents a heatmap of firm transitions between taxable income bins in consecutive years. The horizontal axis denotes the average taxable income bin in year t , and the vertical axis shows the corresponding bin in year $t + 1$. Darker shading indicates a higher number of firms moving between the two bins. The diagonal reflects firms that remain in the same income bin across years, while off-diagonal cells capture upward or downward mobility. Panel (a) illustrates firm movements for 2009–2010, while Panels (b) and (c) plots transitions in 2006–2007 and 2012–2013 for comparison.

Figure A.4: Distribution of Corporate Taxable Incomes by Years and by Sectors



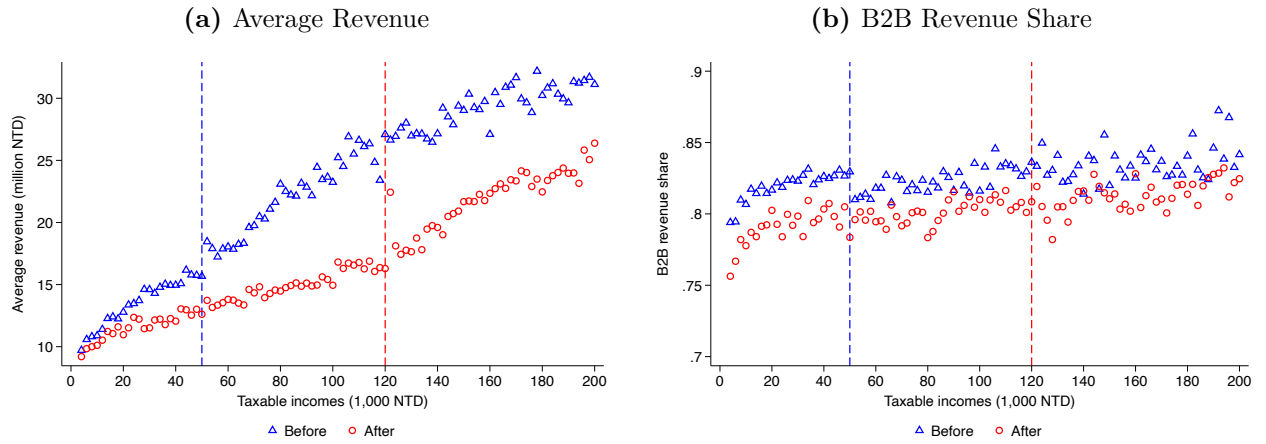
Notes: This figure shows the average annual distribution of corporate taxable incomes across sectors. Panels (a)-(d) plot the average annual distribution of corporate taxable incomes in wholesales, retails, manufacturing, and construction sectors before the reform (2004-2009, in blue triangles) and after (2010-2017, in red circles). The vertical axis denotes the average number of firms yearly in each taxable income bin. The horizontal axis represents the taxable income bins.

Figure A.5: Labor Salary Ratio and Taxable Income: Family and Non-family Employees



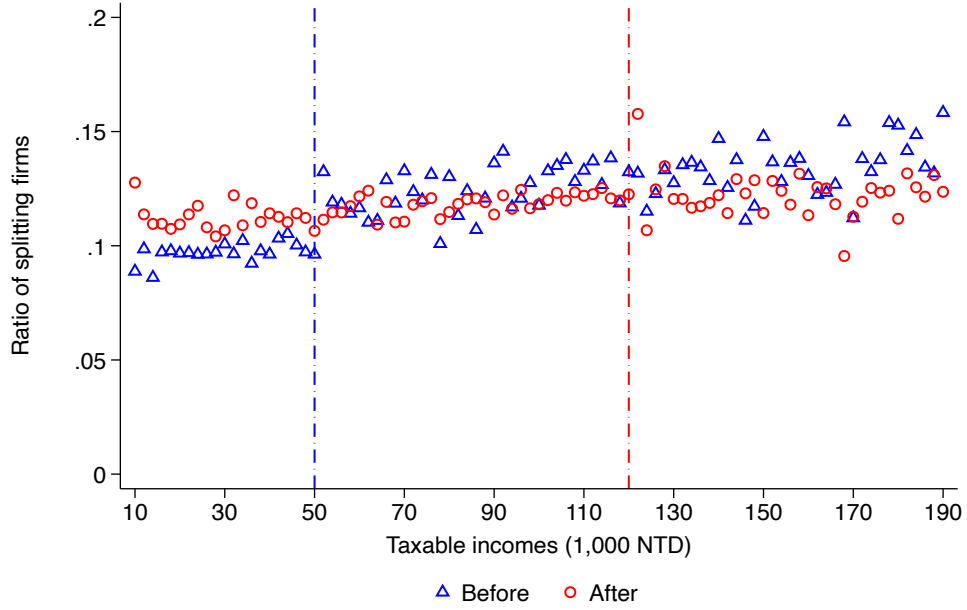
Notes: This figure shows the ratios of labor salary and gross revenue with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the ratios of labor salary paid to family members of firm owners and gross revenue; Panel (b) plots the ratios of labor salary paid to non-family-member employees and gross revenue. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

Figure A.6: How Firms Bunch: Alternative Channels



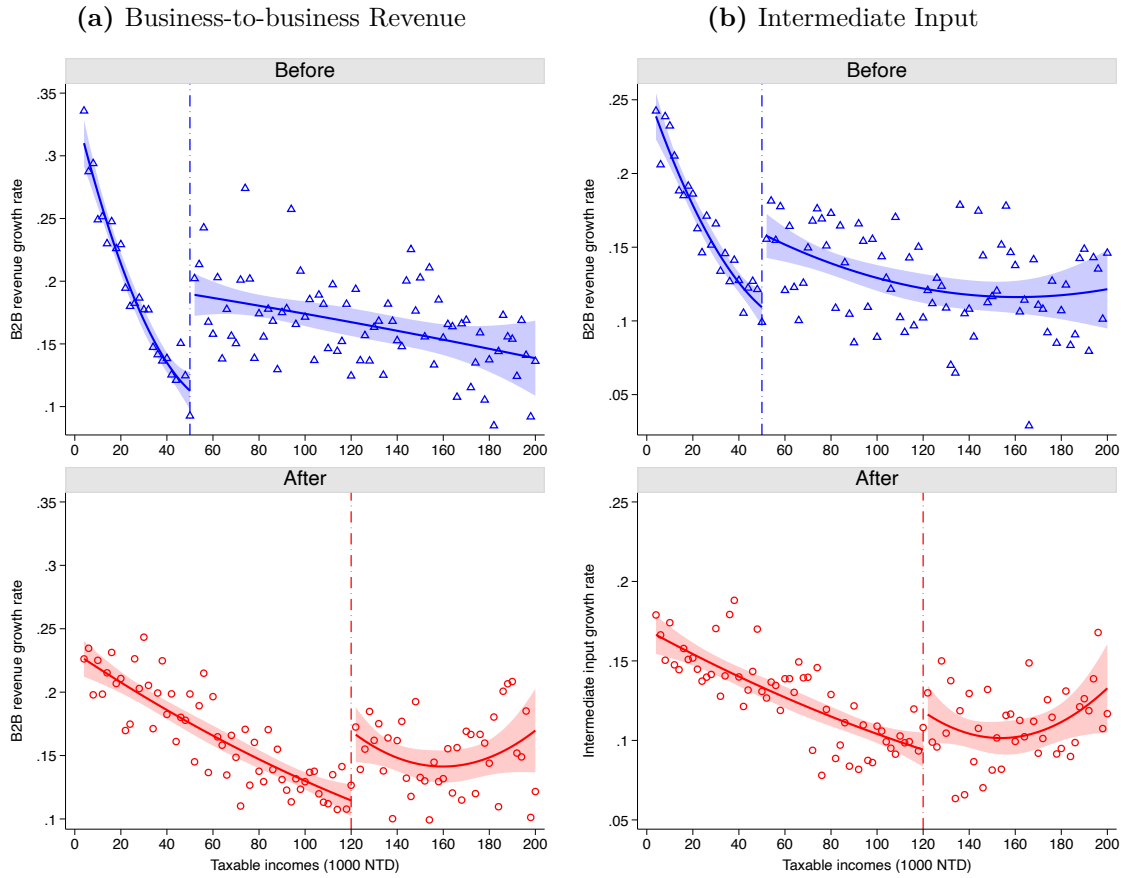
Notes: This figure examines alternative channels for firms to achieve bunching. Panel (a) plots the average firm revenues with respect to corporate taxable income bins before the reform (2004-2009, blue triangles) and after (2010-2017, red circles). Panel (b) plots the shares of business-to-business revenues in gross revenues.

Figure A.7: Ratio of Firm Splitting by Taxable Incomes



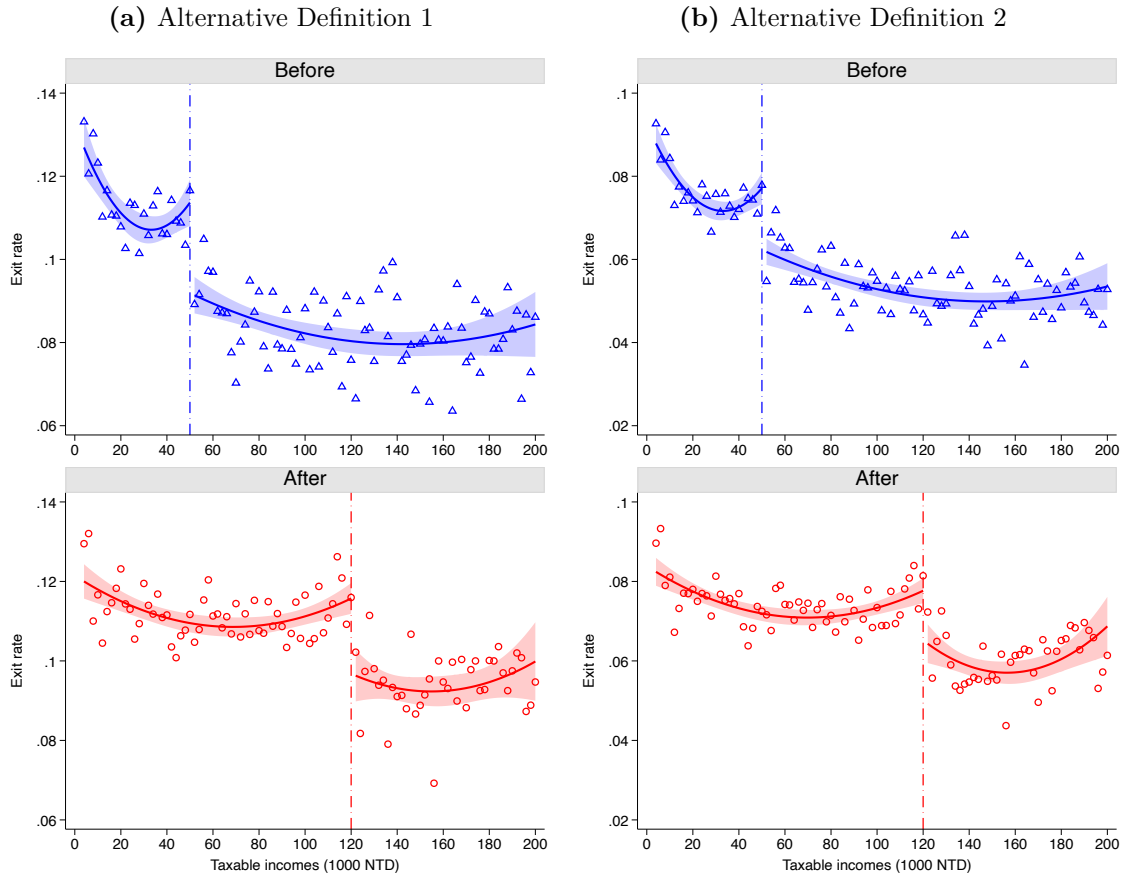
Notes: This figure shows the ratio of splitting firms in each taxable income bin before the reform (2004-2009, in blue circle) and after (2010-2017, in red plus). The vertical axis denotes the ratio of splitting firms to the total number of firms in each taxable income bin. Splitting firms are defined as those which have a common shareholder with any other firms. The horizontal axis represents the taxable income bins. For example, the 5th bin represents firms that have taxable income between 50 thousand NTD (included) and 60 thousand NTD (not included).

Figure A.8: Business-to-business Revenue Growth Rate and Taxable Income



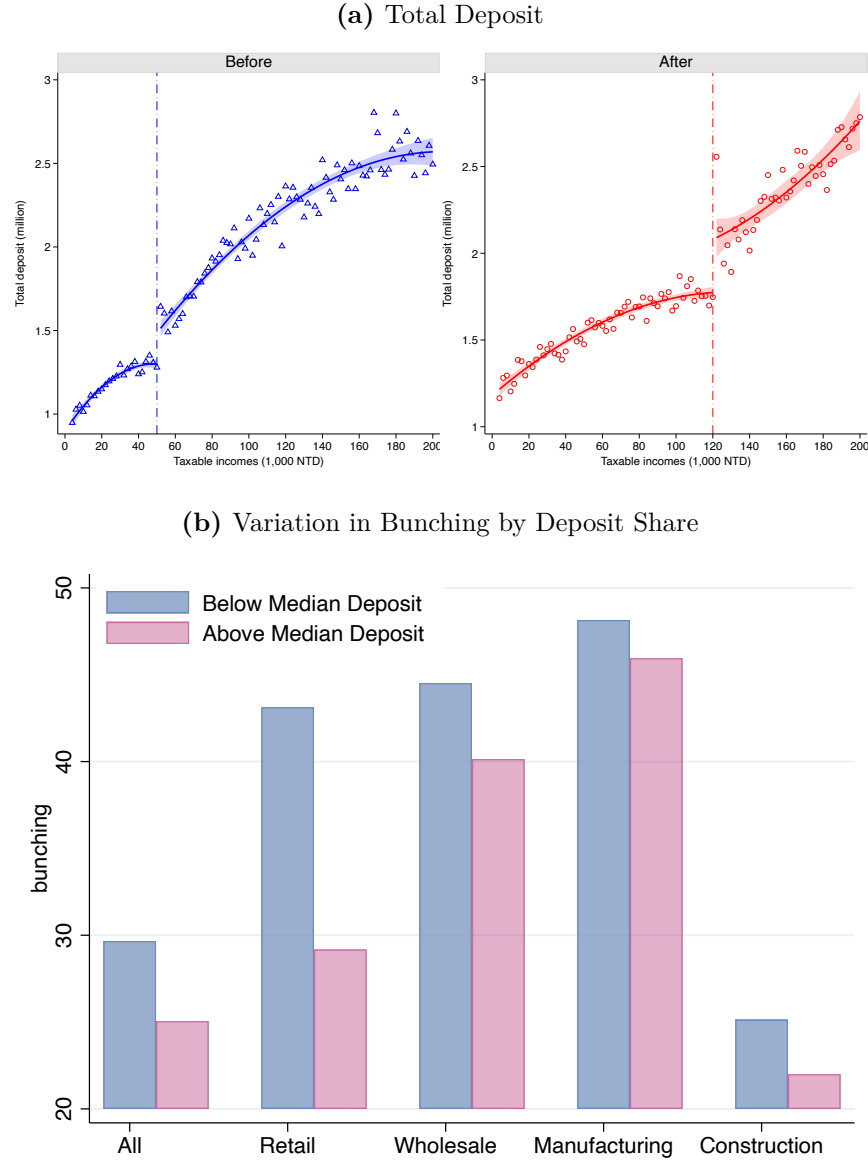
Notes: This figure shows the firm growth rate with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the growth rate of firm revenues from business-to-business transactions; Panel (b) plots the growth rate of firm intermediate inputs. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

Figure A.9: Exit Rate and Taxable Income: Robustness Checks



Notes: This figure shows the average exit rates with respect to corporate taxable income bins before the reform (2004-2009, upper panels with blue triangles and lines) and after (2010-2017, lower panels with red circles and lines). Panel (a) plots the exit rate where firm exit is defined as missing reported tax records; Panel (b) plots the exit rate where firm exit is defined as absent from the tax records for three consecutive years. The horizontal axis represents the taxable income bins. The vertical dashed line indicates the corporate income tax threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

Figure A.10: Liquidity Constraints and Taxable Income



Notes: This figure shows how total deposits vary with taxable incomes. Panel (a) plots total deposits with respect to corporate taxable income bins before the reform (2004-2009) and after (2010-2017). The horizontal axis represents the taxable income bins. The vertical axis denotes the firms' total deposits. The vertical dashed line indicates the corporate income tax exemption threshold. The quadratic fitted lines with 95% confidence intervals are added for taxable income bins below and above the threshold respectively.

B Bunching Estimators

We adopt the bunching estimator following [Saez \(2010\)](#) and [Kleven and Waseem \(2013\)](#) to quantify firms' bunching response to the tax notches after the reform.¹⁵ We first group the data into bins of the firms' corporate taxable income x_j , and estimate the counterfactual distribution using the following equation.

$$f_j = \sum_{i=0}^q \beta_i \cdot (x_j)^i + \sum_{k=x_{ub}}^{x_{lb}} \gamma_k \cdot \mathbb{1}(x_j = k) + \eta_j. \quad (4)$$

f_j is the number of firms in taxable income bin j . We use q to denote the order of the polynomials. By assuming that only firms within the exclusion region $[x_{ub}, x_{lb}]$ would respond, we can use the distribution outside of the exclusion region to construct the counterfactual. γ_k captures the intercept shift generated by firms' response to the policy of each bin within the exclusion region. The counterfactual distribution, in the absence of the policy, is predicted by the follows.

$$\hat{f}_j = \sum_{i=0}^q \hat{\beta}_i \cdot (x_j)^i.$$

We denote the tax threshold as $\bar{x}$ and the bunching mass area (X_B) and the missing mass area (X_H) are given by the follows.

$$\hat{X}_B = \sum_{j=x_{lb}}^{\bar{x}} (f_j - \hat{f}_j) \geq 0, \quad \hat{X}_H = \sum_{j=\bar{x}}^{x_{ub}} (\hat{f}_j - f_j) \geq 0.$$

The estimates of the bunching response are shown in Figure [B.1](#). The solid line represents the estimated counterfactual distribution. The vertical gray line indicates the lower and upper bound of the exclusion region, generated by a data-driven procedure to ensure that the aggregated density of observed and counterfactual distribution is the same. Firms' bunching response dI is defined as the percentage of the firms' density in the exclusive region, which measure the firms average taxable income response to the exemption threshold.¹⁶

¹⁵As [Bertanha, McCallum, and Seegert \(2023\)](#) emphasize, classical bunching estimators depend critically on assumptions about the counterfactual (non-bunching) density outside the excluded region. Because our pre-reform taxable income data are missing or under-represented on the left side of the threshold, extrapolating the density into that region would be unreliable. Hence, to avoid drawing misleading conclusions from such sensitivity, we do not apply the standard bunching estimator to pre-reform observations.

¹⁶Following the bunching literature ([Saez, 2010](#); [Chetty, Friedman, Olsen, and Pistaferri, 2011](#); [Kleven and Waseem, 2013](#); [Almunia and Lopez-Rodriguez, 2018](#)), the bunching estimator b is interpreted as the average response of bunchers. Specifically, $X_B = \int_{\bar{x}}^{x_{ub}} \hat{f}(\bar{x}) d\bar{x} \approx \frac{1}{2} [\hat{f}(\bar{x}) + \hat{f}(x_{ub})] dx$, where $\hat{f}(x)$ represents the counterfactual density of firms' taxable income without the tax notch. Consequently, $dx \equiv \frac{B}{\frac{1}{2} [\hat{f}(\bar{x}) + \hat{f}(\bar{x} + dy^H)]} \approx dx$.

$$dx = \frac{\hat{X}_B}{\frac{1}{2}(\hat{f}_{\bar{x}} + \hat{f}_{x_{ub}})}$$

$\hat{f}_{\bar{x}}$ and $\hat{f}_{x_{ub}}$ are the number firm in each bin of the counterfactual distribution at the tax notch and at the upper bound of the bunching interval, respectively. Following (Chen, Liu, Suárez Serrato, and Xu, 2021; Diamond and Persson, 2016), we adopt a data-driven procedure to jointly select the excluded region (x_{lb}, x_{ub}) and polynomial degree p via K-fold cross-validation. For each candidate triple (p, x_{lb}, x_{ub}) , we fit model 4 on the training fold and evaluate performance on the testing fold using a custom loss function. The loss has two components: (i) $(|X_B/X_H| - 1)^2$, which tests the hypothesis that the excess bunching mass (X_B) to the left of the threshold equals the missing mass (X_H) to the right within the exclusion window; and (ii) the sum of squared prediction errors outside the excluded region. We then select the parameter triple that minimizes this loss function across folds. We apply this procedure separately to the pooled data for the pre-reform and post-reform periods. For the yearly estimator, due to the limited number of observations in individual years, we use the same polynomial degree p and lower bound parameter (p, D_m) as selected from the pooled sample.

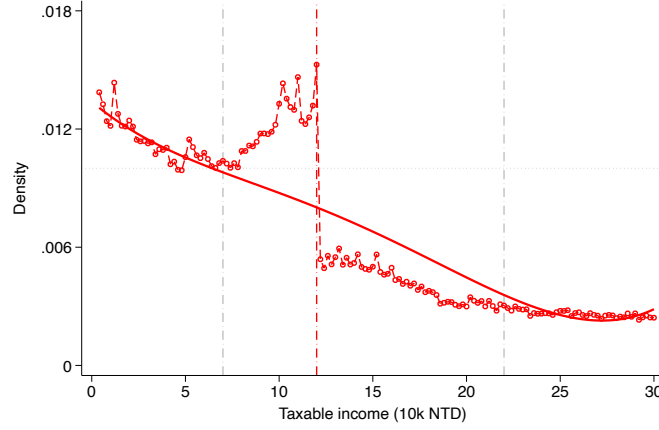
Table B.2 reports bunching estimates for the pooled post-reform period as well as for each year separately. On average, firms reduce reported taxable income by about 27 thousand NTD to avoid exceeding the 120 thousand NTD exemption threshold, which corresponds to roughly 22% of the threshold level. The yearly estimates are broadly stable, showing no clear dynamic pattern either immediately after the reform or in subsequent years, which suggests a persistent behavioral response over the entire post-reform period with no systematic buildup or decline over time.

One thing worth noting is that the bunching estimation is constrained by the fraction of firms in the range $[x_{th}, x_{ub}]$ that do not respond to the threshold, as firms with corporate taxable income above the tax threshold and below the upper bound of the exclusive region do not respond to the threshold. The fraction of firms that do not respond to exempt threshold is between 59-69%.¹⁷

Once we introduce heterogeneity in firms productivity that determine their taxable income, the bunching estimator $dx \approx E(dx)$ is interpreted as weight average of bunch's response and the lack of reaction of non-bunchers.

¹⁷Following the literature, the fraction of firms that do not respond is defined as $a \equiv \frac{\int_{\bar{x}}^{x_{ub}} f_0(x)dx}{\int_{\bar{x}}^{x_{ub}} f(x)dx}$.

Figure B.1: Estimated Counterfactual Density and the Average Bunching Response



Notes: This figure shows the density distribution of corporate taxable incomes after (2010-2017) with the bunching estimator reported at the right corner of each panel. The horizontal axis represents the taxable income bins. The vertical axis denotes the firms' density of corporate taxable income between 0 and 300,000 NTD. The vertical red indicates the CIT exempt threshold before and after the reform. The vertical dashed line represents the lower bound and upper bound of the exclusive region.

Table B.2: Bunching Estimator by Year

	dx	SD	x_{lb}	x_{ub}	a	X_B/X_H	N
Pooled	27.39***	-1.16	80	246	0.76	-1.09	327,289
2010	24.85***	-3.32	80	268	0.80	-0.99	40,226
2011	26.99***	-2.38	80	256	0.77	-1.05	40,222
2012	29.02***	-2.04	80	252	0.75	-1.04	38,678
2013	27.53***	-1.41	80	240	0.74	-1.03	36,393
2014	32.27***	-3.77	80	262	0.74	-1.02	36,006
2015	26.23***	-1.47	80	240	0.75	-0.98	34,531
2016	26.76***	-1.64	80	250	0.77	-1.04	33,929
2017	27.78***	-1.97	80	254	0.76	-0.96	33,472
2018	25.73***	-1.38	80	234	0.75	-1.03	33,832

Notes: Bootstrapped standard errors are shown below each bunching estimator. X_B is the number of firms difference between observed distribution and estimated distribution in the bunching region $[x_{lb}, \bar{x}]$; X_H is the number of firms difference between observed distribution and estimated distribution in the range $[\bar{x}, x_{ub}]$; X_B/X_H is their ratio. x_{ub} and x_{lb} are the bounds of the excluded region. a is the fraction of firms that do not respond to the exempt threshold in $[\bar{x}, x_{ub}]$. N is the number of observations included in the estimation (firms with corporate taxable incomes between 0 and 300 thousand NTD).

C Alternative Empirical Strategy

We exploit a difference-in-differences strategy to examine how tax notches shapes firms' labor and intermediate input decisions. Our baseline specification is presented below.

$$y_{ijt} = \beta \text{Treated}_j \times \text{Post}_t + \lambda_{jt} + \eta_j + \epsilon_{ijt}. \quad (5)$$

We use y_{ijt} to denote the outcome variable for firm i in industry j and year t , including average input cost shares in gross revenue. The dummy variable Treated_j equals one for firms whose taxable incomes were below 50 thousand NTD in 2009 (i.e., the treatment group was subject to the exemption threshold) and zero for firms with taxable income between 50 thousand and 80 thousand NTD in 2009, which serve as a control group. The indicator Post_t takes the value of one for years after the reform (2010 and after) and zero otherwise. The coefficient β on the interaction term $\text{Treated}_j \times \text{Post}_t$ thus captures the average treatment effect of lifting the corporate exemption threshold on firms' labor salary share and intermediate input share.

We control for industry-by-year fixed effects, λ_{jt} , which absorb common macroeconomic shocks that vary across industries or over time, such as aggregate demand fluctuations, inflation, and labor market conditions. Moreover, we include firm fixed effects, η_j , to control for time-invariant firm characteristics such as managerial quality or production technology. Together, these controls mitigate potential confounding factors such as the lingering effects of the 2008 global financial crisis and subsequent recovery to ensure that the estimated β reflects the causal impact of the 2010 corporate income tax reform.

Our key identification assumption is that conditional on industry-by-year fixed effects and firm fixed effects, the treated and control group should follow the same trend in input shares had there not been a change in the exemption threshold. Since we rely on within-firm variation over time to identify the policy impacts, we focus on the static outcome variables such as the labor cost share and intermediate input ratios. The same empirical framework is not well suited to study firms' dynamic responses, as we cannot control for firm fixed effects when analyzing exit decisions.

To bolster the confidence that our estimation uncovers the causal effects, we further conduct an event study to examine the parallel-trend assumption and the dynamic effects of the reform. The empirical model is shown in equation (6). Compared with equation (5), we replace Post_t with a set of dummy variables $\mathbb{I}(t - 2010 = k)$, which is defined as whether year t is k years after 2010. We choose 2009, one year before the reform, as the baseline year. Therefore, β_k measures how the outcome changes compared to 2009 due to the lift of

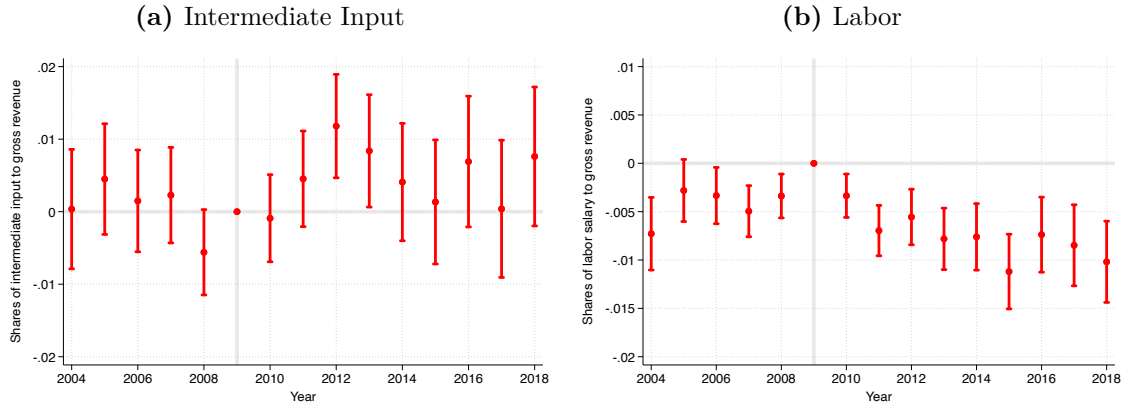
the exemption threshold.

$$y_{ijt} = \sum_{k=-3, k \neq -1}^{k=4} \beta_k \cdot \text{Treated}_j \times \mathbb{I}(t - 2010 = k) + \lambda_{jt} + \eta_j + \epsilon_{ijt}. \quad (6)$$

Table C.1 presents the difference-in-differences estimates. We find that the removal of the 50 thousand NTD exemption threshold leads to an increase in the intermediate input share of revenue and a decrease in the labor salary share. These results suggest that firms engage in bunching by reducing actual production while over-reporting labor expenses.

Figure C.1 displays the estimated β_k coefficients from equation (6), showing the dynamic effect of lifting the exemption threshold on intermediate input share in panel (a) and labor cost share in panel (b). Two key insights emerge from this analysis. First, labor input adjusts immediately, whereas intermediate inputs respond with a lag. This pattern is consistent with the view that reported labor costs can be more easily manipulated for reporting purposes, while intermediate inputs reflect real production decisions that require more time to change. Second, both outcomes display parallel pre-trends prior to the reform, supporting the validity of the parallel trends assumption in our empirical design.

Figure C.1: The Effect of Tax Notches on Input Cost Shares: Event Study



Notes: These figures display the estimated coefficients β_k from the dynamic difference-in-difference equation (6). β_k is the coefficient of interest, which measures the firms' intermediate inputs and labor salary share relative to the reference year. The solid point denotes the point estimates. The line indicates the 95% confidence interval. The horizontal axis represents the number of years after the 2010 corporate income tax reform.

Table C.1: The Effect of Tax Notches on Input Cost Shares: DID

	(1)	(2)	(3)	(4)
<i>Panel A: Intermediate input share</i>		<u>Intermediate input share</u>		
Treatment	0.014*** (0.003)	0.013*** (0.003)	0.014*** (0.003)	0.005* (0.002)
# of Observations	160,066	160,066	150,734	148,329
<i>Panel B: Labor salary share</i>		<u>Labor salary share</u>		
Treatment	-0.008*** (0.002)	-0.008*** (0.002)	-0.009*** (0.002)	-0.004*** (0.001)
# of Observations	159,454	159,454	151,050	148,586
Basic DID controls	✓	✓	✓	✓
Year fixed effect		✓		
Industry-year linear trends			✓	✓
Firm fixed effects				✓

Notes: This table reports the estimation results using empirical model in equation (5). The dependent variable is firms' intermediate input share (Panel A) and labor salary share (Panel B). Column (1) follows the basic DID setup, with variables $Post_t$ and $Treated_i$. Column (2) additionally includes year fixed effects. Column (3) further considers industry-year linear-time trends. Column (4) includes firm fixed effects. Standard errors are clustered at the firm level in parentheses. *p<0.10; **p<0.05; ***p<0.01.